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# TEAM : 21

# AMI1411
# Project Phase I

## DATE : 31 - Aug - 2026

## TEAM MEMBERS :

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# PROBLEM DEFINITION & MOTIVATION

- Patent searching is **time-consuming and complex**, especially when analyzing large patent collections.
- Traditional **keyword-based search** may fail when different terminology is used to describe similar inventions.
- Existing similarity systems often provide a **similarity score without explaining which technical features or claim elements are similar**.
- Manually comparing patent claims, prior-art documents, and technical differences requires **significant expert effort**.
- There is a need for an **AI-assisted, explainable approach** that can understand the semantic meaning of patent claims.

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# OBJECTIVE

- **Identify semantically similar patents** by understanding the technical meaning beyond keyword matching.
- **Decompose patent claims into technical elements** and map corresponding elements across similar patents.
- **Combine semantic embeddings with keyword-based retrieval** to improve the accuracy and relevance of patent search.
- **Analyze prior-art relationships and potential claim overlaps** using AI-assisted comparison and chronological evidence.
- **Generate explainable, evidence-grounded reports** highlighting similarities, differences, supporting evidence, and potential intellectual-property risks.

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# LITERATURE SURVEY

| No. | Paper Title & Year | Authors | Merits | Demerits | Innovation in Proposed System |
|---|---|---|---|---|---|
| 1 | A Semantic Query Expansion-Based Patent Retrieval Approach (2013) | Feng Wang, Lanfen Lin, Shuai Yang, Xiaowei Zhu. | Improves patent retrieval using domain-specific semantic query expansion. | Mainly focuses on query expansion; does not provide detailed claim-level comparison. | Semantic retrieval + terminology understanding to identify patents beyond exact keyword matches. |
| 2 | Comparison and Analysis of Embedding Methods for Patent Documents (2021) | Arousha Haghighian Roudsari, Jafar Afshar, Suan Lee, Wookey Lee | Evaluates different embedding techniques. | Evaluation focuses mainly on representation/classification; does not provide an integrated patent comparison system. | Patent-specific semantic embeddings + similarity retrieval for finding technically related patents. |
| 3 | Patent Citation Network Simplification and Similarity Evaluation Based on Technological Inheritance (2023) | Zhipeng Qiu, Zheng Wang | Combines patent claims, TF-IDF vectors and citation relationships to evaluate technological similarity and inheritance. | Relies mainly on traditional vector representation and citation networks; limited semantic/LLM-based explanation. | Claim similarity + citation/relationship analysis + prior-art timeline in one workflow. |

# LITERATURE SURVEY

| No. | Paper Title & Year | Authors | Merits | Demerits | Innovation in Proposed System |
|---|---|---|---|---|---|
| **4** | Augmented Intelligence for State-of-the-Art Patent Search (2022) | Ana Hafner, Nadja Damij, Dolores Modic | Demonstrates the potential of AI and semantic search for improving patent prior-art searching compared with Boolean keyword search. | Focuses on supporting search rather than detailed claim-element mapping and evidence-grounded reasoning. | Hybrid keyword + semantic retrieval followed by claim-level analysis. |
| **5** | Artificial Intelligence Reducing the Intricacies of Patent Prior Art Search (2023) | Poonam Rawat, Kapil Joshi, Kanishka Vaish, Rajesh Singh, Samta Kathuria, Aditya Verma | Shows how AI can assist keyword suggestion, document retrieval, ranking and visualization to reduce manual prior-art search effort. | Discusses AI-assisted prior-art search but does not provide a complete claim-mapping and explainable LLM platform. | Automated prior-art analysis + evidence-based LLM explanation for faster patent review. |

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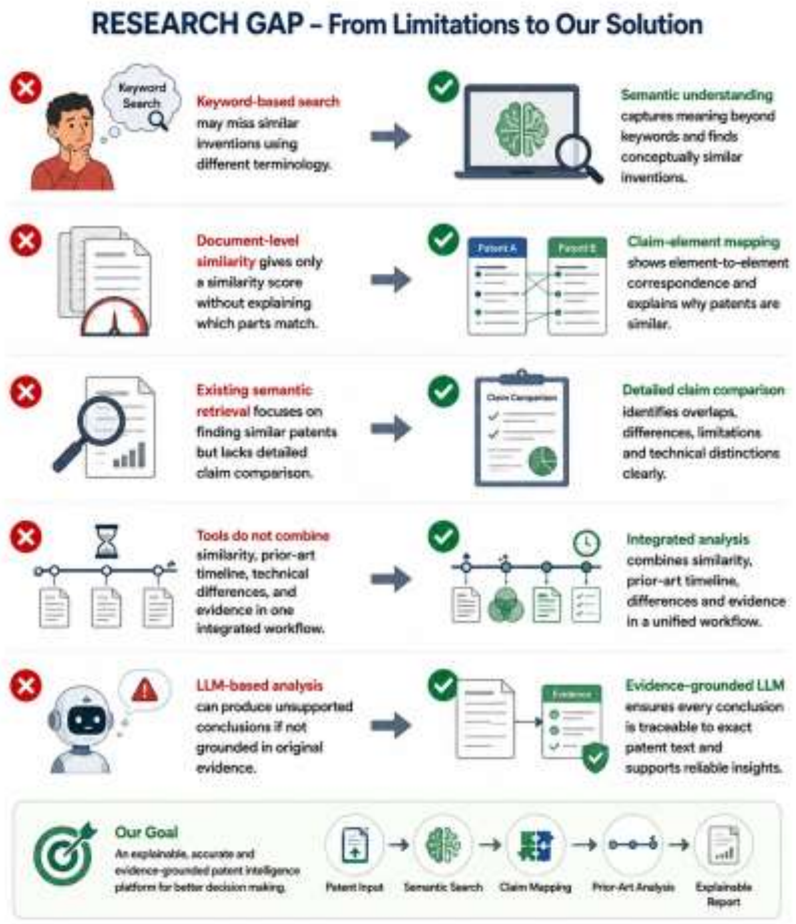
# RESEARCH GAP




- **Keyword-based patent search** may miss semantically similar inventions described using different technical terminology.
- **Document-level similarity models** often provide a similarity score without showing **which claim elements actually overlap**.
- Existing semantic retrieval approaches generally focus on **finding similar patents**, but provide limited support for detailed **claim-to-claim comparison**.
- Patent analysis tools do not always combine **similarity, prior-art chronology, technical differences, and supporting evidence** in one workflow.
- **LLM-based analysis can produce unsupported conclusions** if the generated explanation is not grounded in the original patent evidence.

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# EXISTING SYSTEM

- Semantic equivalents can still be difficult to identify comprehensively.
- Similarity results may not clearly show **which claim elements correspond**.
- Users must manually compare **technical features and differences**.
- Search, prior-art timeline and claim analysis are not necessarily presented as **one integrated workflow**.
- AI-generated explanations require **evidence grounding** to avoid unsupported conclusions.

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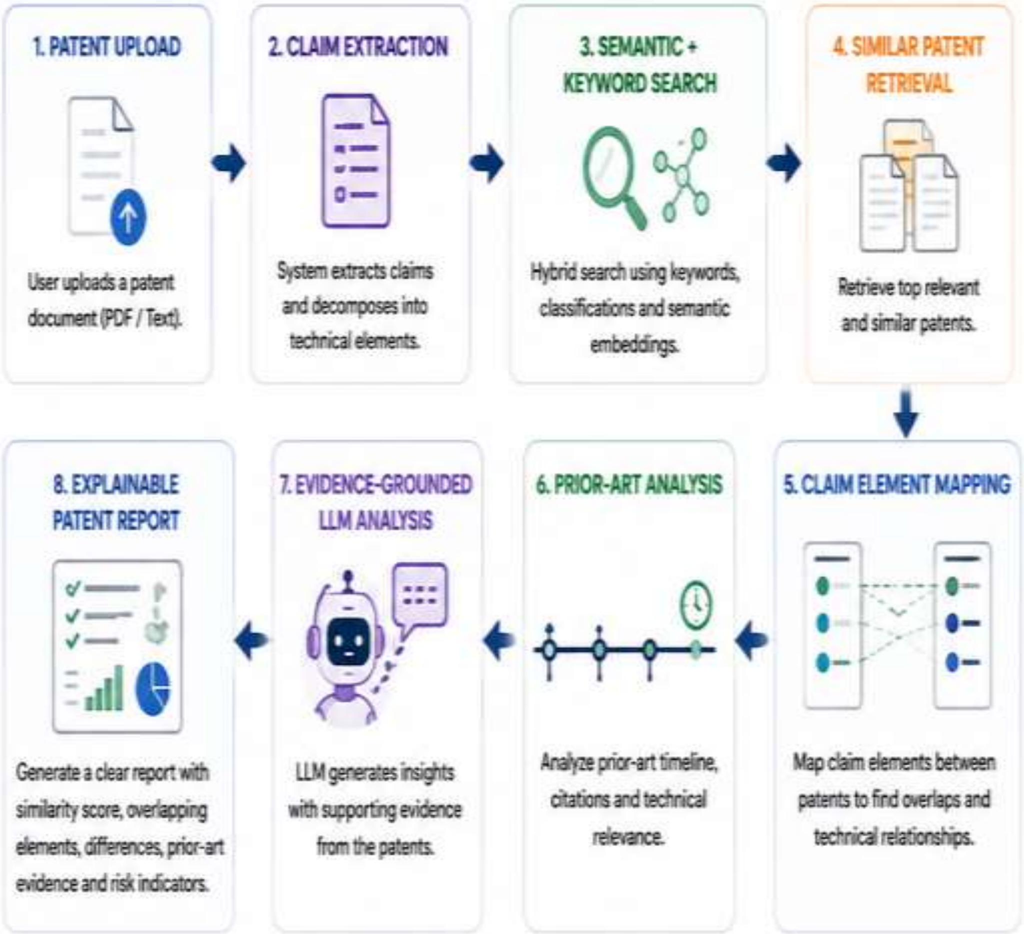
# PROPOSED SYSTEM




- **AI-powered patent analysis platform** that understands the semantic meaning of patent documents beyond exact keywords.
- **Automatically extracts and decomposes patent claims** into meaningful technical elements for detailed comparison.
- **Uses hybrid retrieval** by combining keyword-based search with semantic embeddings to identify highly relevant patents.
- **Maps corresponding claim elements** between patents and identifies similarities, differences, and technical relationships.
- **Uses LLM-based reasoning with supporting patent evidence** to generate an explainable analysis report containing similarity, prior-art evidence, and potential claim-overlap indicators.

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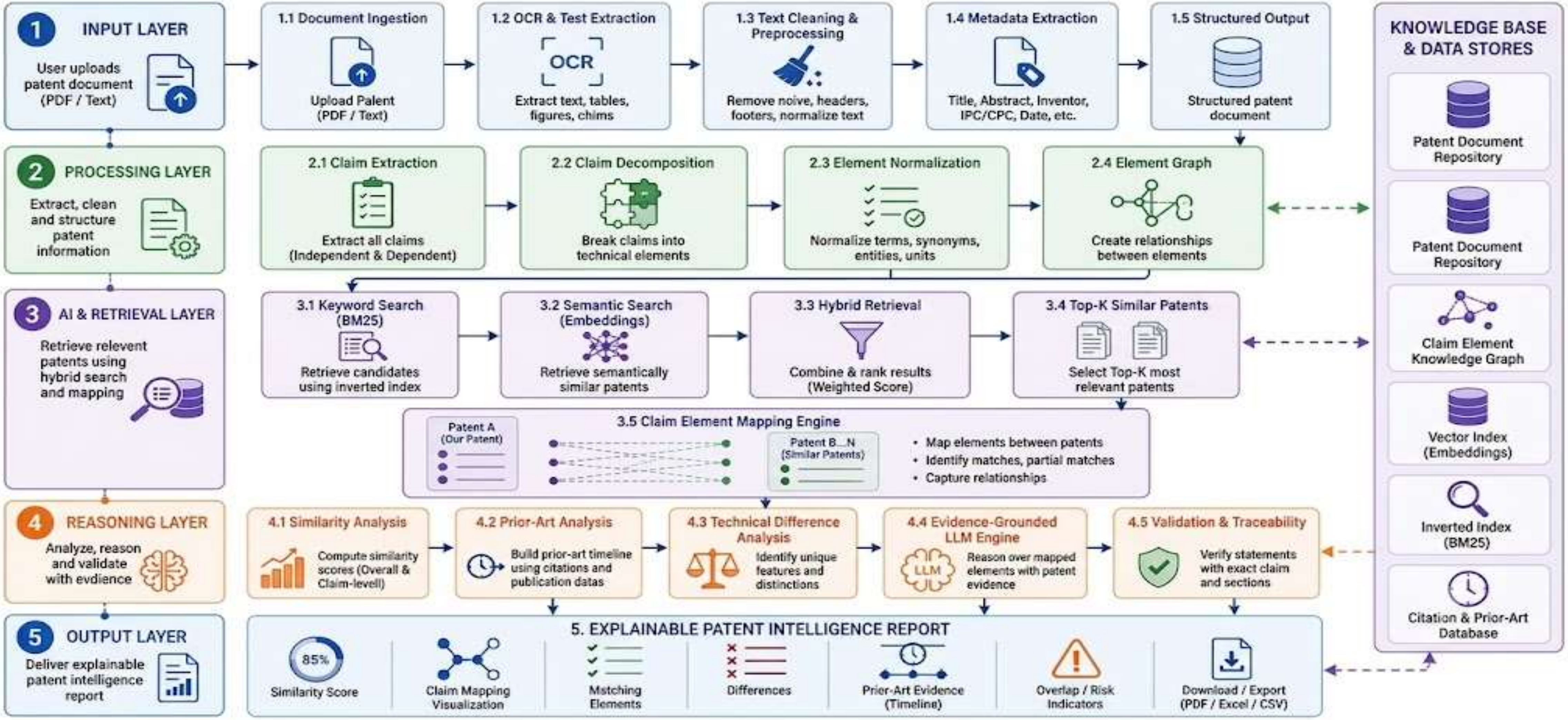
# PROPOSED ARCHITECTURE



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# LIST OF MODULES

- Patent Document Ingestion
- Patent Preprocessing
- Claim Extraction & Decomposition
- Hybrid Patent Retrieval
- LLM-Based Analysis & Reasoning
- Similarity & Risk Analysis
- Explainable Report Generation

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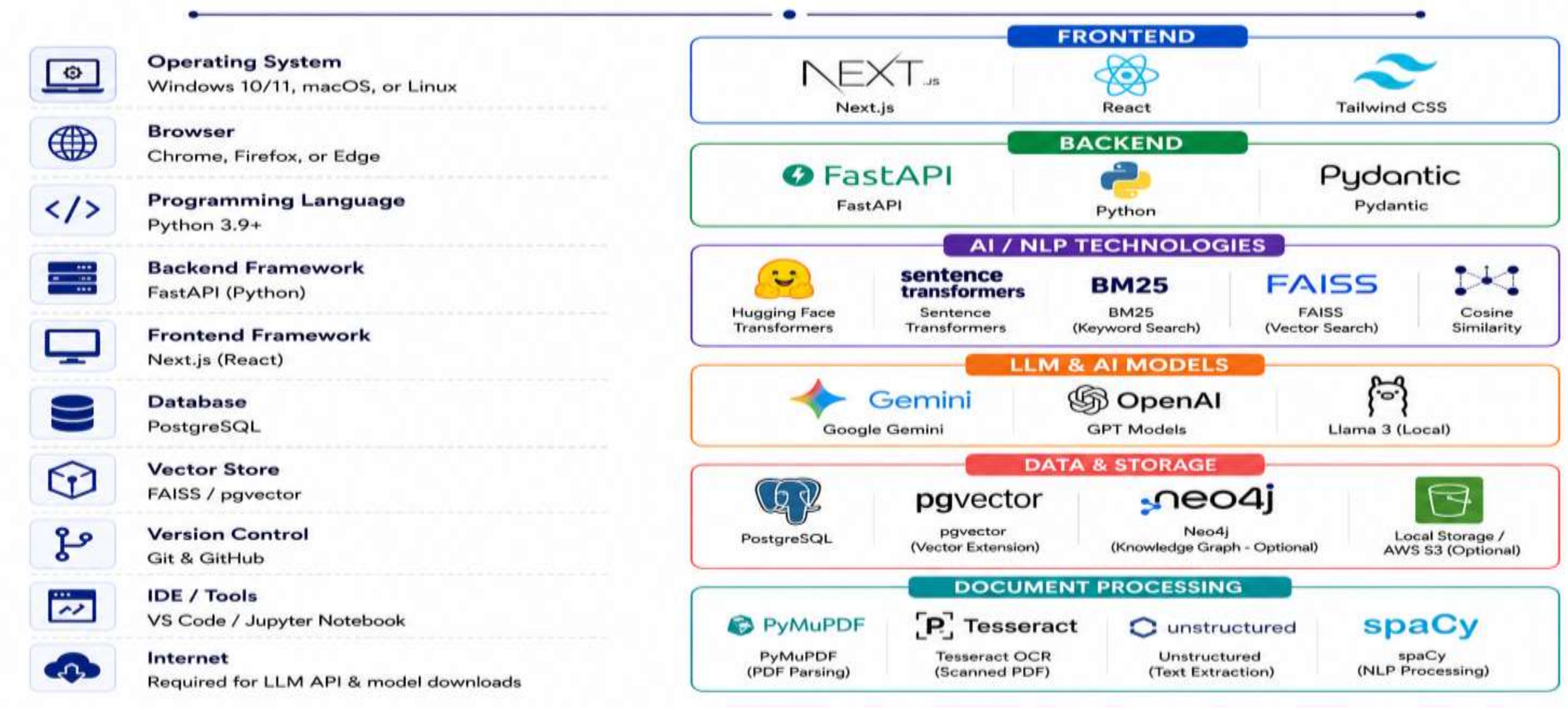
# SYSTEM REQUIREMENTS & TECHNOLOGIES



# INITIAL DESIGN & PROGRESS

- **Patent Upload Interface**

  Upload patent documents in PDF/text format for analysis.

- **Patent Processing Pipeline**

  Designed the workflow for text extraction, cleaning, metadata extraction and claim identification.

- **Claim Analysis Design**

  Designed claim decomposition into individual technical elements.

- **Semantic Retrieval Design**

  Planned hybrid retrieval using **BM25 + semantic embeddings** to identify relevant patents.

- **Explainable Analysis Dashboard**

  Designed the output for claim mapping, similarity scores, prior-art evidence and technical differences.

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# TIMELINE

| PHASE | ACTIVITIES | KEY DELIVERABLES | STATUS |
|---|---|---|---|
| Discovery & Planning | • Market Analysis<br>• Resource Allocation<br>• Scope Definition | • Project Charter<br>• Resource Plan<br>• Market Report | COMPLETED |
| Design & Architecture | • UI/UX Design<br>• System Architecture<br>• Technical Specs | • Wireframes<br>• Prototypes<br>• Architecture Doc | COMPLETED |
| Development | • Front-end Coding<br>• Back-end Engineering<br>• Database Setup | • Core Functionality<br>• API Docs<br>• Codebase | IN PROGRESS |
| Testing & QA | • Unit Testing<br>• System Integration<br>• Bug Fixing & QA | • Test Cases<br>• QA Report<br>• Bug Logs | IN PROGRESS |
| Deployment | • 5.1 Environment Setup<br>• 5.2 Application Launch<br>• 5.3 System Go-Live | • Deployed App<br>• Live System<br>• Release Notes | PLANNED |
| Training & Adoption | • 6.1 User Training<br>• 6.2 Training Materials<br>• 6.3 Adoption Strategy | • Training Manuals<br>• User Feedback<br>• Adoption Report | PLANNED |
| Post-Launch Support | • 7.1 Performance Monitoring<br>• 7.2 Maintenance Updates<br>• 7.3 User Support | • Support Log<br>• Maintenance Plan<br>• Feedback Summary | PLANNED |

# EXPECTED OUTCOME

- **Semantic Patent Retrieval**

  Identify patents with similar inventions even when different terminology is used.
- **Claim-Level Comparison**

  Break claims into technical elements and identify corresponding elements between patents.
- **Prior-Art Intelligence**

  Identify relevant earlier patents and present their relationships through a chronological timeline.
- **Explainable AI Analysis**

  Provide clear explanations of **similarities, differences, and potential claim overlaps** rather than only a similarity score.
- **Automated Patent Intelligence Report**

  Generate a structured report containing similarity scores, claim mapping, prior-art evidence, differences, and potential overlap indicators.

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